



Invited Review

The description of mate choice

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Mate choice is an important evolutionary process influencing a vast array of traits and ecological processes. Although the study of mate choice has proved to be hugely popular, the number of ways in which mate choice can be described is complex and a bewildering array of terminology has developed. The author begins by summarizing some examples of the range of terms used to describe choice that expose this complexity. The author then shows how the information conveyed by different mate choice descriptors can be better understood by comparison to null expectations, that is, the expected variation in a trait when mate choice is not expressed. This comparison is important because many traits that might be affected by mate choice, such as mating rate, mate search effort, and responsiveness, also vary in non-choosy individuals. This is in contrast to other traits, such as the slope of a preference function and mate assessment effort, for which null expectations are predictable. By understanding the null expectation for a trait, its utility as a descriptor of mate choice can be gauged. From this basis, the author suggests an alternative approach to the description of mate choice based upon a principle of describing variation in both “what” is preferred and “by how much” it is preferred. Crucially, the author describes how this approach might apply to a wide range of preference function shapes, thus aiding comparisons across taxa. Finally, the author considers how an improved appreciation of the way mate choice is described can inform future research.

Key words: acceptance threshold, choosiness, mate choice, mating rate, preference, sexual selection.

INTRODUCTION

Mate choice is an important evolutionary process that contributes to selection for a vast array of traits in all manner of color, shape, size, sound, and smell (Andersson 1994). The study of mate choice is hugely popular. The topic “mate choice” has received almost 8000 citations in the Science Citation Index, of which between 500 and 600 citations have been added annually since 2008. However, the variety of ways in which different aspects of mate choice are described, measured, and modelled has developed into an etymological minefield. In this review I begin by presenting an overview of the range of terminology used to describe mate choice. By drawing upon the commonalities and inconsistencies between different approaches, I reveal how mate choice language has developed. I then consider how an improved understanding of mate choice could be gleaned if the description of choice is considered in relation to the null expectations of choice, that is, expected variation in a trait when mate choice is not expressed. I focus, in particular, upon the null expectation for mating rate, as a proposed negative relationship between mate choice and mating rate has been fundamental to many descriptions and perceptions of choice. Through these insights, I suggest an alternative approach to the description of choice that encompasses the most important aspects of variation in choice and which is comparable across a wide range

of preference functions. I conclude by discussing how an improved understanding of the description of mate choice can contribute to more informed research and the realization of novel opportunities.

A DEFINITION OF MATE CHOICE

Mate choice can be defined as the process that occurs whenever the effects of traits expressed in one sex leads to non-random matings with members of the opposite sex (Halliday 1983; Kokko et al. 2003). This broad definition of choice encompasses both direct and indirect forms of mate choice (sensu Wiley and Poston 1996). In addition to the decision to mate with specific individuals, mate choice is also expressed when there is variation in the amount of resources invested in specific mates (e.g., Bonduriansky 2001). For example, an individual could bias the allocation of sperm (e.g., Engqvist and Sauer 2002) or provide greater parental care (e.g., Sheldon 2000) to specific mates. Mate choice can therefore be more inclusively defined as the process that occurs whenever the effects of traits expressed in one sex leads to non-random allocation of reproductive investment with members of the opposite sex. Within this definition the decision to mate can itself be considered an act of reproductive investment.

MATE CHOICE TERMINOLOGY

There is a plethora of terms used to describe different aspects of, and variation in, mate choice (Table 1; also see Box 1 in Widemo

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Table 1
Examples of diversity in the language used to describe choice

Definition	Term	Example reference
The relationship between a phenotypic trait in potential mates (<i>x</i> axis) and the reproductive resources invested in a mate (<i>y</i> axis).	Preference function	Lande (1981)
The order in which potential mates are ranked.	Preference function	Jennions and Petrie (1997)
The shape of a preference function, e.g., linear, stabilizing, disruptive (Figure 1a–e).	Preference form	Cotton et al. (2006)
	Preference shape	Brooks and Endler (2001)
	Choosiness	Jennions and Petrie (1997)
The amount of resources invested into choice, such as mate search effort and mate assessment effort.		
A step preference function in which potential mates with trait values greater than the threshold are accepted and all others are rejected (Figure 1a).	Acceptance threshold	Janetos (1980)
Variation in the slope of a preference function—in general reference (Figure 1f–i)	Choosiness	Kokko and Mappes (2013)
	Choosiness	Ratterman et al. (2014)
	Preference strength	Cotton et al. (2006)
	Preference score	Dosen and Montgomerie (2004)
Variation in the slope of a preference function—in reference to a categorical preference function (Figure 1f)	Preference	Lande (1981)
Variation in the slope of a preference function—in reference to a directional preference function (Figure 1g)	Preference slope	Brooks and Endler (2001)
	Sensitivity	Rowe et al. (2005)
	Discrimination	Ritchie et al. (2005)
	Tightness	Ritchie (1996)
Variation in the slope of a preference function—in reference to a stabilizing preference function (Figure 1h)		
Width of a stabilizing preference function – analogous to the slope (Figure 1h).	Tolerance	Rodriguez et al. (2013b)
The location of a peak of a stabilizing preference function (Figure 1m)	Peak preference	Rodriguez et al. (2013a)
	Preference	Lande (1981)
Average response to potential mates (Figure 1o–r)	Responsiveness (an aspect of choosiness)	Brooks and Endler (2001)
	Preference score	Tinghitella et al. (2013)
	Receptivity	Lynch et al. (2005)
Variation in the response to potential mates (Figure 2a–f compared with Figure 2c–h).	Discrimination (an aspect of choosiness)	Brooks and Endler (2001)
	Selectivity	Hedrick and Weber (1998)
	Preference strength	Rodriguez et al. (2013b)
	Permissiveness	Lynch et al. (2005)
A response to a signal that is normally unattractive.	Selectivity	Fowler-Finn and Rodriguez (2013)
A composite of responsiveness, tolerance, and preference shape that describes how a response declines as signals deviate from peak preference.		
Expressing a consistent response to different stimuli in repeated choice trials.	Discrimination	Lynch et al. (2005)

and Sæther (1999) and Figure 1 in Cotton et al. (2006)). This diversity is in part justified as there are multiple aspects of mate choice that can be described (Jennions and Petrie 1997) and because a single suite of terms may not be appropriate for all study systems (Bonduriansky 2001). Nevertheless, the variation in terms used can appear bewildering. Here, I present an overview of some prominent terminology and different, though non-mutually exclusive, approaches to the description of mate choice.

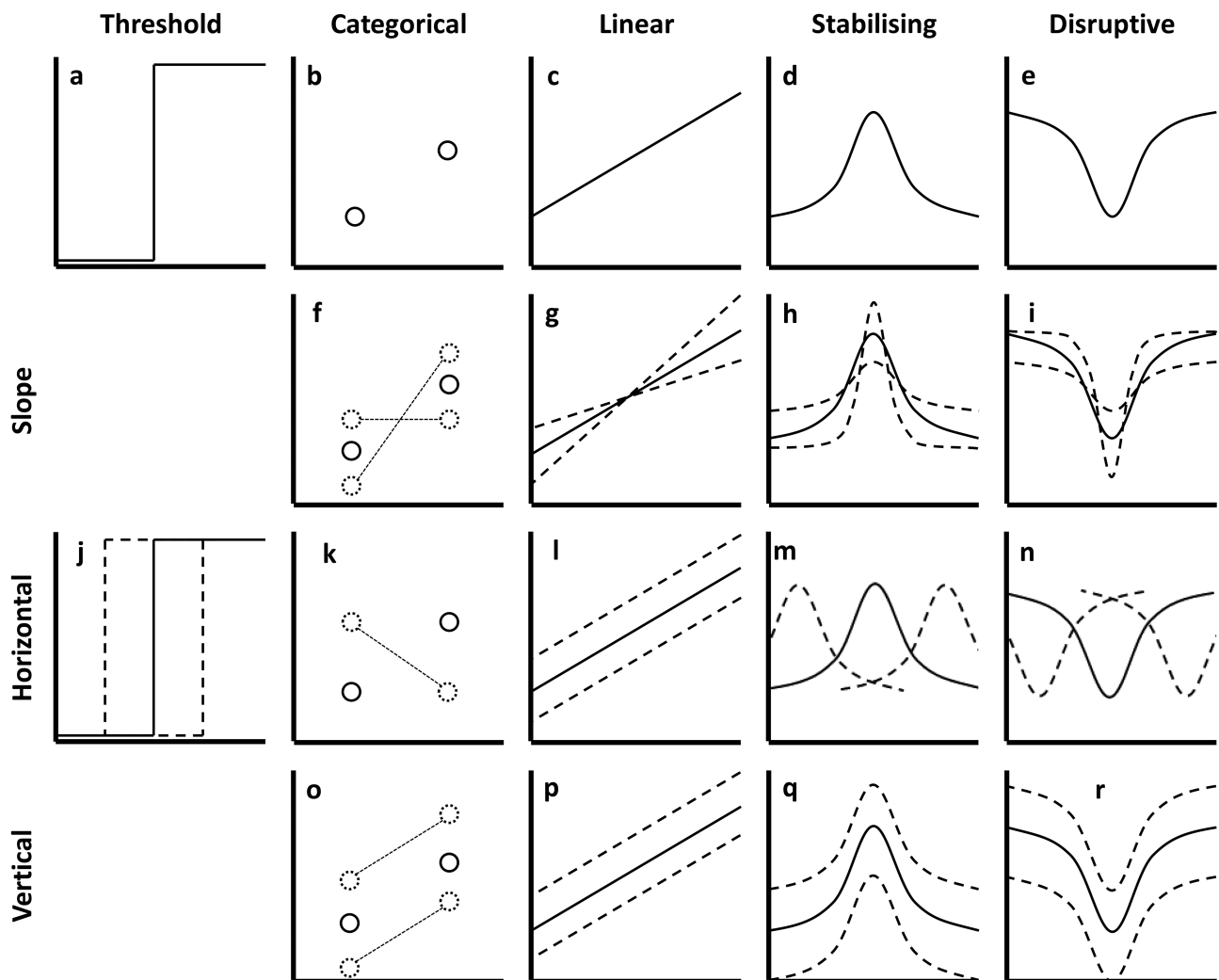
Preference functions

The concept of preference functions, and the term “preference function” have been widely and consistently used to describe patterns of mate choice (e.g., Basolo 1998; Gerhardt et al. 2000; Jennions and Petrie 1997; Ritchie 1996; Wagner 1998; Figure 1). Preference functions have been particularly influential in the study of mate choice as the concept spans both empirical and theoretical approaches. The mathematical interpretation of preference functions has been widely used to model variation in choice, of which many examples can be traced back to the influential work of figures such as Lande (e.g., Lande 1981). On the *y* axis of a preference function is a variable describing the amount of resources invested in reproduction with each mate (Bonduriansky 2001). This variable is often the probability of an individual mating though many other traits can also usefully represent actual reproductive investment or a willingness to invest in reproduction with a mate, for example, association time, courtship intensity or duration, latency to mate,

number of sperm ejaculated, and parental care (e.g., Hoefler et al. 2009; Kelly and Jennions 2011; Matessi et al. 2009; South et al. 2012).

On the *x* axis of a preference function is a phenotypic trait expressed by potential mates. In addition to continuous phenotypic variation, the *x* axis can also depict categorical traits, including continuous traits reduced to categories (e.g., Crowley et al. 1991; Fisher and Rosenthal 2006; Hubbell and Johnson 1987; Figure 1b), or indeed relative, rather than absolute, variation in a trait. Furthermore, phenotypic variation in potential mates can often be complex in nature, and this can be depicted in multivariate preference functions (e.g., Backwell and Passmore 1996; Brooks et al. 2005; Candolin 2003). In summary, preference functions are incredibly useful for the description of choice because a wide range of preferred traits and expressions of choice can be depicted within the same framework.

When viewing preference functions, a preference is shown whenever variation in reproductive investment (*y* axis) is dependent upon phenotypic variation in potential mates (*x* axis). A preference function then describes a preference whenever the function is not flat and, importantly, the absence of a preference when the function is flat. Variation in mate choice is then demonstrated not by variation in the intercept of a preference function, but variation in the slope of a preference function under different conditions. That is, when the relation between reproductive investment (*y* axis) and phenotypic variation in potential mates (*x* axis) is dependent on a further

**Figure 1**

Variation in preference functions. Each of the 18 preference functions depicts investment of resources in potential mates (y axis) versus phenotypic variation in potential mates (x axis). Five different “shapes” of preference function are depicted in each of the columns. The first row is a reference example of each preference function shape – Threshold, Categorical, Linear, Stabilizing and Disruptive. Row 2 (f–i) depicts variation in the slope, row 3 (j–n) depicts variation in the horizontal position, and row 4 (o–r) depicts variation in the vertical position of each preference function. Solid lines and circles show the reference example and dashed lines and circles show variation in each preference function. Column 1 (a and j) depict a threshold preference function where the y axis is the probability of mating. All potential mates with trait values greater than the acceptance threshold are mated, whilst all potential mates with trait values less than the acceptance threshold are not mated. This preference function can only vary in the value of the acceptance threshold, that is, horizontal variation (j). There is no variation in the slope as this is a step function and no variation in vertical position as the y axis is bounded between 0 and 1. Column 2 (b, f, k, o) depicts a categorical preference function where phenotypic variation in potential mates falls into 2 distinct classes. Horizontal variation in the position of this preference function is viewed as a switch in the direction of the x axis without variation in the slope or vertical position (k). Dotted lines (f, k, and o) show connected elements of the same preference function.

variable, for example, environmental conditions or age, generating a significant interaction term (e.g., Fisher and Rosenthal 2006; Qvarnström et al. 2000).

Acceptance thresholds

As a description of variation in mate choice, the acceptance threshold (also termed “choice threshold” or simply “threshold”) came to prominence following the work of Janetos (1980) and was later refined, most notably, by Real (1990) and Parker (1983). The acceptance threshold model of mate choice is typically represented by a “step” preference function whereby all potential mates with trait values less than an acceptance threshold are rejected and all potential mates with trait values greater than an acceptance threshold

are accepted (Figure 1a). Alongside acceptance thresholds, “best-of- n ” mate choice strategies have also been described (Janetos 1980). In essence, acceptance threshold and “best-of- n ” models of mate choice both involve a step preference function. A crucial difference is that for an acceptance threshold strategy the threshold is an internally defined, fixed property of the chooser, whilst for a “best-of- n ” strategy an optimal threshold is determined following a period of mate searching to compare prospective mates. The relative success of each strategy is therefore dependent on the costs of mate searching and assessment (Real 1990; Wiegmann et al. 1996).

Acceptance thresholds have been widely adopted to describe mate choice, for example, to quantify the strength of mate choice or as a component of choosiness (e.g., Jennions and Petrie 1997;

Kokko and Mappes 2013; Milner et al. 2010; Parker 1983; Real 1990; Zhou et al. 2011). Likewise, many empirical studies have sought to determine the value of the acceptance threshold in various taxa (e.g., Beckers and Wagner 2011; Cooley and Marshall 2004; Dugatkin 1998; Forsgren 1997; Parri et al. 1997). Correlating acceptance thresholds with the strength of choice, or choosiness, can make intuitive sense as individuals with higher acceptance thresholds are selecting a smaller “elite” portion of potential mates. Individuals with higher acceptance thresholds will also need to expend greater mate search effort to find a sufficient number of acceptable mates (e.g., Jennions and Petrie 1997).

The acceptance threshold has been an important and influential concept in the study of mate choice. However, an important limitation of describing mate choice solely in terms of an acceptance threshold is that this does not incorporate variation in the slope of a preference function, that is, choice is modelled as a step function rather than continuous variation in the allocation of reproductive investment (Valone et al. 1996; Figure 1j compared with Figure 1f–i). Indeed, a strict step function may rarely be expressed, particularly as this would require perfect information about potential mates. Alternatively, a threshold can describe the inflection point of a continuous function that also varies in slope (e.g., Forsgren 1997; Rowe et al. 2005). Therefore, depending on the questions being asked, considering acceptance thresholds alongside other aspects of a preference function can reveal a more complete and accurate depiction of choice (Rowe et al. 2005).

Distinguishing preference and choosiness

A commonly adopted framework for describing mate choice is to distinguish preference from choosiness (e.g., Cotton et al. 2006; Jennions and Petrie 1997; Widemo and Sæther 1999). This demarcation is important as it differentiates innate tendencies toward specific mates (i.e., preferences – choices that would be made if cost was no object) and the actual mating bias that results depending on the amount of effort that a choosing individual is willing or able to invest in mate choice (i.e., choosiness; Cotton et al. 2006).

Jennions and Petrie (1997) provide a detailed description of terminology in which mate choice is defined as the pattern of mating that arises from “mating preferences.” “Mating preferences” are further divided into “preference functions” and “choosiness.” “Preference functions” are defined as the order in which an individual ranks potential mates. In contrast, “choosiness” is the amount of resources invested into choice, principally mate search effort and mate assessment effort. As mate search effort is coupled with variation in acceptance thresholds (i.e., individuals with higher thresholds will need to expend more effort searching for mates; see Acceptance thresholds above), variation in acceptance thresholds is positively correlated with “choosiness” in this definition (Jennions and Petrie 1997).

In an approach echoing that of Jennions and Petrie (1997), Cotton et al. (2006) expanded the description of preference functions to further define the “form” and “strength” of a preference function. The “form” of a preference function can take many shapes, for example, directional, stabilizing, or disruptive (Figure 1a–e) and is thus analogous to the ranking of potential mates. For a positive directional preference function larger trait values are ranked highest (and vice versa), for a stabilizing preference function intermediate trait values are ranked highest, and for a disruptive preference function extreme trait values are ranked highest. In addition, Cotton et al. (2006) defined the “strength” of

a preference function as the rate of change, or slope, of the preference function, that is, how much higher an individual is ranked for a given phenotypic difference (e.g., Figure 1f–i).

Responsiveness and discrimination

Another approach to the description of “choosiness” has been to describe choosiness as an outcome of “responsiveness” and “discrimination” (Bailey 2008, 2011; Brooks 2002; Brooks and Endler 2001; Ritchie et al. 2005). “Responsiveness” can be defined as a measure of motivation to mate, or the mean response of a focal individual to potential mates, and can thus be represented by a wide variety of traits, for example, courtship intensity, response latency, or association time. Variation in responsiveness can be depicted as a vertical shift in the position of the preference function (Figure 1o–r and see also Bailey 2008). In contrast, “discrimination” has been defined as the variation in response to different individuals. This can be calculated, for example, as the standard deviation of all responses (Brooks and Endler 2001) or the difference between a response to the most preferred stimulus and the average response to all stimuli (Gray and Cade 1999), though either method can yield similar results (Bailey 2008).

The slope of a preference function

The slope of a preference function can be defined as the difference in reproductive resources invested, including the likelihood of mating, per unit change in trait value of potential mates (e.g., Murphy and Gerhardt 2000). The slope of a preference function can be described for both linear and non-linear preferences through regression coefficients (e.g., Basolo 1998; Hunt et al. 2005; Murphy and Gerhardt 2000; Wagner et al. 1995; Figure 1g–i). The slope of a preference function can also be described for categorical traits as the difference in resources invested in potential mates belonging to different classes (e.g., Fisher and Rosenthal 2006; Qvarnström et al. 2000; Tinghitella et al. 2013; Figure 1f).

The slope of a preference function has been widely used to describe mate choice and so the importance of this trait is clearly appreciated. Crucially, whenever the slope of a preference function is flat then choice is not expressed. This means that the value of the slope is a fundamental aspect of preference functions that defines whether mate choice is expressed. Consequently, models that vary the slope of a preference function (e.g., Kirkpatrick 1996; Lande 1981; Pomiankowski et al. 1991) have taken a fundamentally different approach when contrasted with acceptance threshold models of mate choice in which the slope of the preference function is typically invariant (though also see Rowe et al. (2005) for an example in which both threshold and slope are varied). Terminology used to describe the slope of a preference function can vary however (Table 1) and, notably, “choosiness” has been used in reference to both the slope and threshold of a preference function. This can make it difficult to draw accurate comparisons between different studies.

Summary of mate choice terminology

Multiple aspects of mating behavior can be used to describe mate choice, each of which providing distinct information about variation in choice and whose relevance may depend on the specific question being asked. However, the distinction between different components of choice and the information that each confers can become obscured when inconsistent terminology is used. For example, “choosiness” has been used on different occasions to refer

to the effort invested into mate choice, the acceptance threshold, the slope of a preference function, and responsiveness (Brooks and Endler 2001; Jennions and Petrie 1997; Kokko and Mappes 2013; Ratterman et al. 2014; Table 1). Though within individual studies a term may be clearly and consistently defined, a unified framework for the description of variation in choice is notably lacking. Consequently, when comparing studies it is important to recognize that the same terminology may have been used to describe different aspects of choice, or the same aspect of choice may be described using different terms.

NULL EXPECTATIONS

It is illustrated above how the terminology used to describe mate choice is diverse and this has generated a complex web of definitions that can be confusing to navigate. Nevertheless, this complexity reveals that mate choice cannot be described with a single metric. Preference functions can vary in many different respects (Figure 1) and an individual can vary the amount and type of resources invested in mate choice (Jennions and Petrie 1997). However, in addition to variation in the mating behavior of choosy individuals, it is also important to recognize that the mating behavior of non-choosy individuals can vary too. The crucial difference between choosy and non-choosy individuals is whether any variation in mating behavior is biased toward specific mates. Consequently, to determine whether variation in a particular behavior reflects variation in mate choice, it is important to recognize the patterns of variation that are expected when there is no choice – the null expectation. Below, I consider the appropriate null expectations for a range of different traits to gain insight into the information conveyed by each trait that is relevant to mate choice.

Mate choice and mating rate

There is a common perception that the mating rate of choosy individuals will be reduced compared with non-choosy individuals (e.g., Kokko and Mappes 2013; Widemo and Sæther 1999). The principle that mate choice causes a reduction in mating rate has been explicitly incorporated into some models of mate choice. For example, where choosy individuals have been defined as mating with a specific subset of potential mates, non-choosy individuals have been defined as mating with all potential mates (e.g., Crowley et al. 1991; Hubbell and Johnson 1987; Kirkpatrick 1982; Parker 1983). The numerous models of mate choice that simulate variation in the acceptance threshold of a preference function also reinforce a perception that mate choice causes a reduction in mating rate (e.g., Bleu et al. 2012; Gavrilets et al. 2001; Houle and Kondrashov 2002; Hutchinson and Halupka 2004; Johnstone et al. 1996; Kokko and Mappes 2005, 2013; Owens and Thompson 1994; Seubert et al. 2011; Wiegmann et al. 1999). This is because the number of potential mates that are acceptable will decrease as the acceptance threshold is increased.

Perceived negative associations between mate choice and mating rate have also influenced the measurement of choice in empirical studies. For example, Tinghitella et al. (2013) based their measure of mate choice on the supposition that, “A female who is not choosy would attempt to spawn with every male encountered....” Choice has elsewhere been measured as the number of potential mates that are rejected without also recording phenotypic variation among accepted and rejected individuals (e.g., Lenton and Francesconi 2011; Narraway et al. 2010).

However, the overwhelming focus on a negative association between mate choice and mating rate overlooks that non-choosy individuals do not have to mate with all potential mates (e.g., Reynolds and Gross 1990). As long as the subset of potential mates that are mated is a random sample of all potential mates then no choice is expressed, irrespective of how many potential mates are accepted. Alternately put, as long as the preference function is flat then, whatever the value of the intercept, there is no choice. This means that there is no fixed null expectation for mating rate in the absence of choice. As there is no fixed null expectation, it cannot be assumed that choosy individuals will always mate less often than their non-choosy counterparts.

Mating rate and acceptance thresholds

If a fixed null expectation for mating rate, and thus a negative association between mate choice and mating rate, cannot be assumed, this would appear to contradict the negative correlation between mate choice and mating rate that can arise when choice is modelled as an acceptance threshold. It is therefore worth considering how variation in an acceptance threshold influences mating rate. An acceptance threshold effectively reduces a continuously varying trait to a dichotomous variable, with all the individuals in one class, and none in the other class, being mated. The value of the threshold determines where the cut off is, thus how many potential mates are in each class. This ultimately influences mating rate. However, variation in the threshold does not influence by how much one class of potential mate is preferred over the other. This remains a constant when the slope of the preference function is modelled as a step function (Valone et al. 1996; Figure 1j). Acceptance threshold models thus contrast with other models of mate choice that vary the slope of a preference function (e.g., Kirkpatrick 1996; Lande 1981; Pomiankowski et al. 1991). Acceptance threshold models can be augmented to include variation in the slope (e.g., Rowe et al. 2005) yet the threshold is still describing what is preferred (i.e., the horizontal position of the inflection point) and not by how much individuals either side of this point are favored (i.e., variation in the slope).

As variation in the acceptance threshold represents a horizontal change in the position of a preference function, this might be compared with, for example, varying the horizontal position of a peaked preference function (i.e., Figure 1j,m; e.g., Murphy and Gerhardt 2000). Thus, we might view horizontal variation in a preference function, either acceptance threshold or peak, as influencing “what” is preferred but not “by how much” it is preferred, which is instead influenced by variation in the slope (Figure 1f–i). This perspective on the information conferred by acceptance thresholds has important implications for interpreting the results of these models. Most importantly, as the threshold does not affect “by how much” something is preferred, choice is never completely absent in individuals that express variation in an acceptance threshold. This is because varying only the value of the acceptance threshold of a step preference function can never produce a truly flat preference function – the null expectation (Figure 1j). In contrast, if variation in choice is defined as the slope of a preference function then choice can be absent whenever the slope is flat.

It might be argued that when an acceptance threshold is lower than the minimum trait value of potential mates, choice is not expressed because all potential mates are accepted. This supposition would be incorrect however. All individuals that express a threshold assess potential mates and judge them against this standard. Even an individual with a low acceptance threshold still

assesses and judges each potential mate. This is irrespective of whether, following the expression of choice, all potential mates turn out to be suitable. This situation is in stark contrast to an individual with a truly flat preference function, who has no criteria for mate acceptance and so will not expend any effort assessing mates. In other words, there is a very significant and important difference between an individual with a low threshold who happens to accept all potential mates because they all happen to satisfy an assessed criteria and an individual without any choice criteria that accepts all mates without any assessment. The importance of this difference can be understood by considering that when an individual with a very low threshold is observed to accept all mates, this observation is dependent upon the range of potential mates presented to the choosing individual. If a different set of potential mates with sufficiently lower trait values were available, then the choosing individual, following the same choice rules, would have rejected some mates. In contrast, a non-choosy individual having a flat preference function with maximal intercept would always be observed to accept all potential mates irrespective of the range of variation in mates that is available. Thus ultimately, even though all potential mates might be accepted when acceptance thresholds are low, this is not evidence that an individual has not expressed choice.

In summary, models that define variation in choice through the acceptance threshold of a preference function simulate choice in a very different way to models that vary the slope of a preference function. This distinction can be summarized as the difference between describing “what” is preferred and “by how much” something is preferred. To interpret the predictions of mate choice models it is important to understand these differences in the way that mate choice variation can be generated. In particular, as varying the acceptance threshold of a preference function cannot result in a flat preference function, studies that vary only this trait lack comparison to a null expectation. It can then be informative to generate a random mating model in parallel to the mate choice model (e.g., Hutchinson and Halupka 2004; Janetos 1980). However, when simulating random mating patterns, it is important to recall that a flat preference function can vary in intercept so that non-choosy individuals may not necessarily mate with all potential mates that are encountered.

Mate search effort and mate assessment effort

Mate search effort is important for determining reproductive success and may influence or reflect patterns of mate choice (e.g., Jennions and Petrie 1997; Parker 1983; Real 1990). For example, additional mate search effort may be necessary to recoup lost mating opportunities if potential mates are rejected in favor of a preferred alternative (Parker 1983). Likewise, an individual with a higher acceptance threshold may need to expend greater search effort to ensure a sufficient number of acceptable mates are found (Jennions and Petrie 1997; Real 1990).

However, non-choosy individuals may also need to search for mates and hence the null expectation for mate search effort is not fixed at zero. In the same way that mating rate can vary in the absence of choice, it can also be expected that non-choosy individuals will vary in mate search effort. Without an understanding of variation in mate search effort when there is no choice, the null expectation, it is difficult to unambiguously correlate variation in mate search effort with variation in mate choice.

This situation in respect of mate search effort is contrasted in mate assessment effort. As non-choosy individuals do not treat potential mates differently there will be no selective

advantage to mate assessment. This generates a clear, fixed null expectation of zero mate assessment effort in non-choosy individuals. Consequently, mate assessment effort can be expected to provide distinct and less ambiguous information about variation in mate choice when compared with mate search effort.

Responsiveness and discrimination

Describing “choosiness” in terms of “responsiveness” and “discrimination” notably differs from other notions of “choosiness” (e.g., Cotton et al. 2006; Jennions and Petrie 1997). Jennions and Petrie (1997) defined choosiness as the amount of resources invested into choice, which can be positively correlated with the acceptance threshold. Choosiness was thus associated with a horizontal change in the position of a preference function (Figure 1j). In contrast, variation in “responsiveness” describes a vertical change in the position of a preference function (Figure 1o–r and Bailey 2008). Furthermore, “discrimination” has been countenanced as a measure of choosiness on the basis that this might reflect variation in the slope of a preference function (Bailey 2008; Brooks and Endler 2001; Gray and Cade 1999). Yet, the slope of a preference function has been associated with “preference,” rather than “choosiness” (Brooks and Endler 2001; Cotton et al. 2006). These inconsistencies raise an important question of precisely how “responsiveness” and “discrimination” reflect variation in mate choice. Some insight can again be achieved through considering the null expectation for these traits.

Alongside varied mating rates and varied mate search effort, non-choosy individuals can vary in motivation toward potential mates (“responsiveness” sensu Brooks and Endler 2001). Variation in responsiveness might arise from variation in, for example, age, condition, or mating status (e.g., Judge et al. 2010; Pachevo et al. 2013). Hence there is no clear, fixed null expectation for responsiveness in non-choosy individuals (e.g., Figure 2). As “responsiveness” can vary in non-choosy individuals then, without understanding this underlying variation, “responsiveness” cannot provide an accurate and unbiased description of mate choice. The crucial difference between choosy and non-choosy individuals can more accurately be described not as a difference in responsiveness, but whether variation in responsiveness is correlated with phenotypic variation in potential mates, that is the slope of a preference function (Figure 1f–i).

Nevertheless, if “responsiveness” does vary because of choice it is unclear what the nature of this variation will be. One expectation may be that a choosy individual will show greater responsiveness to preferred than non-preferred mates, hence responsiveness will correlate with phenotypic variation in potential mates (e.g., Bailey 2008, 2011; Brooks and Endler 2001; Ritchie 1996). Alternatively, responsiveness may reflect the time or effort that an individual invests in mate assessment (e.g., Brooks and Endler 2001) which would be consistent with the definition of “choosiness” proposed by Jennions and Petrie (1997). If responsiveness represents mate assessment effort, choosier individuals would be expected to uniformly increase responsiveness to all potential mates. This is clear because phenotypic variation is ascertained only as part of the assessment process. Consequently, increased responsiveness that does not correlate with phenotypic variation could be a sign of increased choosiness. To illustrate this point, latency to mate has been used as a measure of responsiveness in female fruit flies, *Drosophila melanogaster* (e.g., Narraway et al. 2010). However, when latencies to mate are short this could be because 1) choice is being expressed and preferred males are being mated sooner or 2) the female is expressing

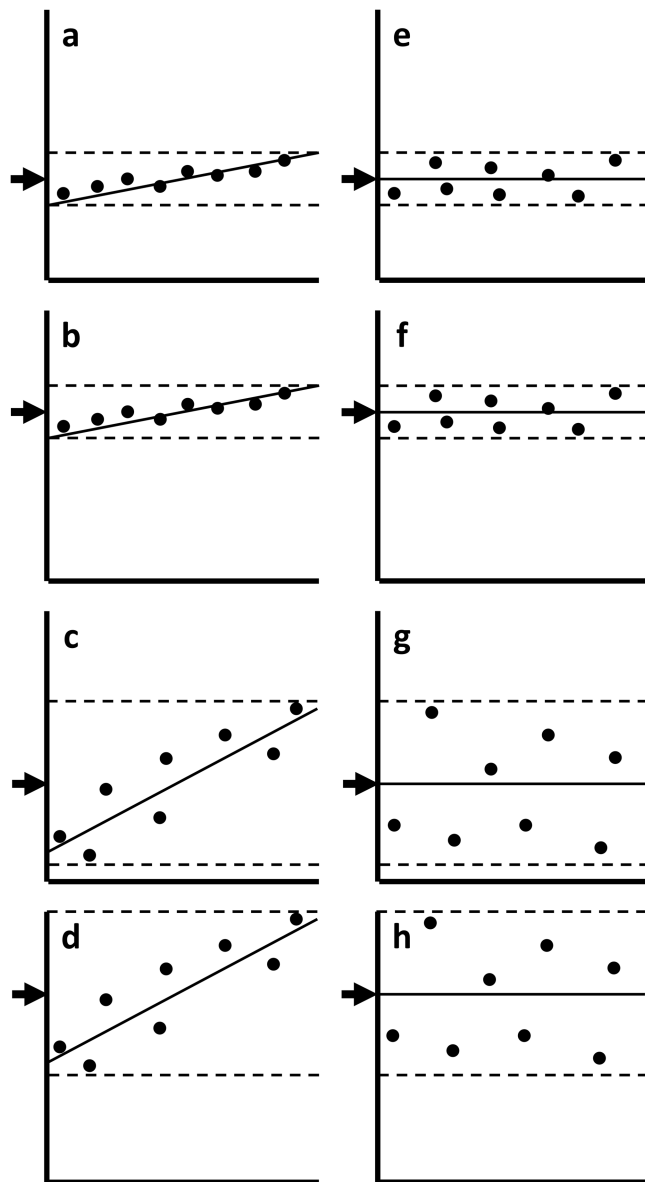


Figure 2 “Responsiveness” and “discrimination” can vary when there is no choice. Each of the 8 preference functions depicts motivation to mate (y axis) versus phenotypic variation in potential mates (x axis). Arrows indicate mean motivation to mate, that is “responsiveness” sensu Brooks and Endler (2001). The distance between dashed lines indicates variation in responsiveness to mates, that is, “discrimination” sensu Brooks and Endler (2001). Plots in the left panel illustrate how variation in responsiveness (a vs. b), discrimination (a vs. c), and both responsiveness and discrimination (a vs. d) can reflect variation in mate choice. Corresponding plots in the right panel (e, f, g, and h) illustrate analogous variation in these traits without any expression of choice.

less choice by spending less time assessing each male before they are mated. Consequently, for responsiveness to be used to describe choice this would require knowledge of the mechanisms involved to determine if the trait used reflects mate assessment effort or reproductive investment following mate assessment.

In addition to “responsiveness” (i.e., differences in the mean response to potential mates), “discrimination” (i.e., variation about the mean response to potential mates; sensu Brooks and Endler 2001) can also differ among non-choosy individuals (Figure 2e–h)."

Hence there is also no clear, fixed null expectation for “discrimination.” The importance of varying “discrimination” in non-choosy individuals can be gleaned from a model of mate choice that varied courtship effort (Servadio and Lande 2006). In this model, non-choosy individuals courted all potential mates with the same intensity, hence discrimination was zero. Meanwhile, choosy individuals courted preferred mates more than average and non-preferred mates less than average, hence discrimination was greater than zero. Choosy individuals thus differed from non-choosy individuals in 2 respects. First, choosy individuals biased allocation of courtship effort to specific mates but second, choosy individuals varied in the amount of courtship offered to potential mates. An alternative no-choice mating strategy that might be considered is if non-choosy individuals varied in how much they court potential mates (discrimination greater than zero) but without biasing this variation toward specific mates (e.g., Figure 2e compared with Figure 2g). Considering alternative mating strategies such as this may be important in interpreting the results of these kinds of models. This is because non-choosy individuals that have a more variable response to potential mates could have a different likelihood of mating when compared with non-choosy individuals that express a consistent response to all potential mates.

Summary of null expectations

When interpreting different behaviors that may indicate variation in mate choice it is insightful to consider the null expectation for each trait. For example, mating rate, mate search effort, “responsiveness,” and “discrimination” can all vary in non-choosy individuals. Without understanding the underlying variation in such traits it is not possible to unambiguously infer that any variation reflects variation in choice. This is in contrast to traits such as mate assessment effort and the slope of a preference function which have clear, fixed null expectations as these traits will be equal to zero in non-choosy individuals. Considering the null expectation can therefore be important in interpreting the information that different traits can provide which is relevant to the description of mate choice.

THE FUTURE DESCRIPTION OF MATE CHOICE

This review highlights a need for future research to adopt a more consistent and unified approach to the description of mate choice. A significant factor that will influence the success of any approach is how easily it can be applied across a range of taxa. Preference functions already provide a solid foundation for this description as each axis can be adapted to describe a wide array of traits depicting mate choice. However, the shape of preference functions can vary significantly (Figure 1a–e) and there are many more shapes that have not been depicted here. Yet, for many different shape of preference function, the principal variation within each can be divided into 1 of 3 types—variation in the slope (Figure 1f–i), horizontal position (Figure 1j–n), and vertical position (Figure 1o–r). It would be beneficial to recognize variation in these analogous components of choice so that comparisons between species or groups of individuals can be made even where the shape of a preference function is different. For example, variation in the slope of a linear preference function and steepness of a peaked preference could be considered as analogous traits (Figure 1g,h). Likewise, variation in the acceptance threshold and peak of a stabilizing preference both represent variation in the horizontal position of a preference function (Figure 1j,m). Acknowledging these shared differences could

significantly help in reducing the array of terminology that is currently relied upon.

It is important to consider the information that each principal aspect of variation (i.e., slope, horizontal position, and vertical position) of a preference function conveys about variation in choice. Because non-choosy individuals are defined as having a flat preference function, the slope and horizontal position of a preference function are invariant for non-choosy individuals. In contrast, the vertical position of a preference function can vary in both choosy and non-choosy individuals alike. Variation in the slope and horizontal position of a preference function can thus be distinguished as providing information that describes variation only present in choosy individuals.

These 2 aspects of variation in a preference function (i.e., slope and horizontal position) each describe a distinct aspect of choice. In layman terms, the horizontal position of a preference function can be viewed as describing “what” is preferred. For example, the location of a peak of a stabilizing preference function is the most preferred trait value and the acceptance threshold describes the range of potential mates that will be accepted. For a categorical trait, variation in the horizontal position of a preference function is equivalent to varying the category of trait that is preferred without varying the slope or vertical position of the function (Figure 1k). In contrast, the slope of a preference function can be viewed as describing “by how much” something is preferred, that is, by how much are preferred mate types favored relative to other mate types. In summary, the principal variation in mate choice can be described through variation in 2 aspects of a preference function regardless of the shape of the preference function – “what” is preferred (the horizontal position) and “by how much” (the slope).

These 2 pieces of information that describe mate choice echo previous approaches to the definition of mate choice that differentiated innate biases an individual might have toward certain mates from the manifestation of those biases (e.g., Cotton et al. 2006; Jennions and Petrie 1997). Consequently, the most appropriate term for referring to any variation in the horizontal position of a preference function would be “preference,” that is, which type of mate does the individual have a “preference” for. This term has previously been used in the same context when describing the location of a peak of a stabilizing preference function (e.g., Lande 1981; Rodriguez et al. 2013b). “Choosiness” might then be used to refer to variation in the slope of a preference function (e.g., Ratterman et al. 2014). An important corollary of defining “choosiness” as the slope of a preference function is that, irrespective of taxa and the shape of a preference function, the absence of choice could universally be described as whenever “choosiness” is equal to zero. In summary, an individual could be described as exhibiting a “preference” for certain mate types (the horizontal position of a preference function), but the extent of those preferences will depend on “choosiness” (the slope of a preference function).

OPPORTUNITIES FOR FUTURE WORK

When and how will choice evolve?

A fundamental question in respect of mate choice has been when it will evolve. Understanding the relationship between mate choices and mating rate is crucial to answer this question as mating rate is fundamental to fitness. Unsurprisingly therefore, the perceived negative association between mate choice and mating rate is a highly cited and significant proposed cost that could hinder the evolution of mate choice (e.g., Bleu et al. 2012; Brown et al. 2009; Kokko

et al. 2003, 2006; Kokko and Johnstone 2002; Kokko and Mappes 2013; Lenton and Francesconi 2011; Owens and Thompson 1994). A reduction in mating rate caused by mate choice could be particularly costly in males as the relation between reproductive fitness and mating rate is generally stronger (Edward and Chapman 2011).

However, this perspective on the relationship between mate choice and mating rate is largely based upon false assumptions that non-choosy individuals will mate with all available mates and a negative association between mate choice and mating rate that arises in acceptance threshold models of mate choice. By considering variation in other components of choice that might not impact mating rate to the same extent as acceptance thresholds, such as the slope of a preference function, it might be found that these aspects of choice have greater freedom to evolve. For example, although “preference” for a rarer mate type can contribute to a reduced mating rate, variation in “by how much” that mate type is favored over others (i.e., “choosiness”) might not influence mating rate to the same extent (e.g., Rowe et al. 2005) and could even have a positive effect on mating rate if there is a preference for common mate types. Thus “choosiness” may evolve more freely than “preferences” due to reduced constraints associated with a reduction in mating rate.

A greater appreciation of these different aspects of choice could contribute to understanding the expression of choosy behavior where it may otherwise be unexpected. Thus instead of asking when choice will evolve, a more pertinent question may be how each aspect of choice will evolve. It could be quite revealing if existing theoretical models that are focussed on a single aspect of variation in choice were developed to incorporate variation in multiple aspects of a preference function. Alternate prospects for the evolution of choice may then become apparent. For example, although an increase in the acceptance threshold is often constrained because of a costly influence on mating rate, variation in the slope of a preference function may not face similar limitations. A prime example of this kind of opportunity is the way in which Rowe et al. (2005) developed the model originally presented by Gavrillets et al. (2001). By including variation in the slope (termed “sensitivity”) of a preference function the expected outcomes were significantly altered and new insight was generated. Rather than increased female choice resulting from sexually antagonistic selection, females were instead able to evolve indifference to male traits. Through developing a better understanding of the different metrics available to describe choice, there are significant opportunities for future research to consider how different components of mate choice evolve, either independently or as co-evolving traits (e.g., Bailey 2008). An improved understanding of different choice metrics and their relation to each other would also ensure that the most appropriate metric(s) to describe choice, relevant to the question being asked, are used in models of mate choice.

Mate choice effort

A key consideration in describing mate choice is the effort that is invested into choice by choosy individuals, principally mate search effort and mate assessment effort (e.g., Jennions and Petrie 1997; Widemo and Sæther 1999). Though these 2 processes are often combined under a banner of “mate sampling effort,” I would argue that a distinction needs to be maintained as the null expectation for each trait is fundamentally different.

A number of studies have sought to determine mate search effort (e.g., Dunn and Whittingham 2007; Naulty et al. 2013) and mate assessment effort (e.g., Backwell and Passmore 1996; Schwartz et al.

2004) though it may often be difficult to differentiate the 2 (e.g., Byers et al. 2005). For example, mate search effort and mate assessment effort can become conflated when individuals are searching for a specific type of mate (e.g., Guevara-Fiore et al. 2010). Mate assessment effort may also be difficult to distinguish if this effort is largely invested in neurological processes that are difficult to observe directly. The observed ability of an individual to differentiate potential mates may also not reliably indicate the effort that has been invested due to individual variation in mate assessment efficiency. The ability to differentiate potential mates with the same accuracy may require varying levels of effort or require different lengths of time depending on the individual (Chittka et al. 2009). The empirical measurement of mate assessment effort, and distinguishing this from mate search effort, therefore presents a significant challenge. However, understanding how individuals invest resources into mate assessment, how this equates to the accuracy of mate assessment, and the relation between mate assessment effort and preference functions is likely to be crucial in understanding patterns of mate choice (Castellano and Cermelli 2011; Chittka et al. 2009).

CONCLUSIONS

As we have seen, the description of mate choice can be complex and a confusing array of terminology has developed in this field. Explanations for this are that the popularity of the subject has generated many different perspectives and there are many aspects of mate choice that can be described. Researchers need to be vigilant in the language used to describe mate choice as it cannot be assumed that the same term used in different instances is describing the same component of choice, nor that different terms will always describe different components of choice. Further, when choosing a trait to describe mate choice, little may be understood about variation in choice if there is no clear expectation or understanding of how the trait will vary when there is no choice. In this review, I propose a more integrated approach to the description of mate choice that distinguishes variation in the horizontal position of a preference function (i.e., preference) from variation in the slope of a preference function (i.e., choosiness). An advantage of this approach to describing mate choice is that it could be applied to a wide range of preference functions that vary in shape, thus allowing greater comparison across taxa. From an improved understanding of how choice is described, significant opportunities for future research to advance our understanding of mate choice are revealed.

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